Quantification: 682_z23
=======================
PRE-VALIDATION: no number here has been checked against manual (ImageJ) counts.
PIXEL UNITS: lengths px, areas px^2, densities px^-1. The images carry no micron calibration.
Statistics are over interior lacunae only (those not touching the frame edge).

Per image
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| measure | value |
|---|---|
| lacuna count | 14 |
| interior lacuna count | 11 |
| frame-edge lacuna count | 3 |

Interior lacunae only: mean, median and sample SD (n = 11)
----------------------------------------------------------

| measure | kind | mean | median | sd | units |
|---|---|---|---|---|---|
| area_px2 | lacuna shape | 1994.91 | 1519 | 1026.72 | px^2 |
| major_axis_length_px | lacuna shape | 85.7455 | 86.1566 | 22.5145 | px |
| minor_axis_length_px | lacuna shape | 30.4546 | 26.8577 | 8.8339 | px |
| aspect_ratio | lacuna shape | 2.9668 | 3.0363 | 0.8704 | unitless |
| eccentricity | lacuna shape | 0.92 | 0.9442 | 0.0643 | unitless |
| solidity | lacuna shape | 0.8555 | 0.8762 | 0.0782 | unitless |
| roots_count | headline | 6.6364 | 6 | 2.0136 | count |
| ring_length_r30_px | headline | 300.091 | 325 | 73.1676 | px |
| ring_length_r60_px | reported | 819.182 | 824 | 137.998 | px |
| owned_length_px | ownership-dependent | 1767.03 | 1286.96 | 1121.22 | px |
| edge_count | ownership-dependent | 79.1818 | 59 | 52.7443 | count |
| mean_edge_length_px | ownership-dependent | 22.8818 | 23.7818 | 2.1918 | px |
| perimeter_px | normalised | 221.572 | 210.244 | 51.1926 | px |
| ring_area_r30_px2 | normalised | 8909.82 | 8646 | 1432.67 | px^2 |
| ring_area_r60_px2 | normalised | 22488.8 | 22056 | 3166.3 | px^2 |
| in_frame_fraction_r30 | normalised | 1 | 1 | 0 | unitless |
| in_frame_fraction_r60 | normalised | 0.9972 | 1 | 0.0094 | unitless |
| ring_density_r30 | normalised | 0.0340089 | 0.0329632 | 0.00927786 | px^-1 |
| ring_density_r60 | normalised | 0.0367476 | 0.0364527 | 0.00652245 | px^-1 |
| roots_per_100px_perimeter | normalised | 3.0582 | 2.8549 | 0.9582 | per 100 px |
| ring_attached_length_r30_px | network v2 | 238.454 | 221 | 84.8167 | px |
| ring_attached_length_r60_px | network v2 | 519.545 | 547 | 170.821 | px |
| ring_length_w_r30_px | network v2 | 337.45 | 363.078 | 81.5864 | px |
| ring_length_w_r60_px | network v2 | 924.825 | 946.879 | 148.656 | px |
| sholl_crossings_r10 | network v2 | 7.5455 | 7 | 2.0181 | count |
| sholl_crossings_r20 | network v2 | 10.7273 | 11 | 2.7601 | count |
| sholl_crossings_r30 | network v2 | 12.4545 | 12 | 3.4457 | count |
| ring_width_mean_r30_px | width | 6.2512 | 6.2404 | 0.4247 | px |

Per field
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| measure | value | units |
|---|---|---|
| analysed_area_px2 | 1.02271e+06 | px^2 |
| total_skeleton_length_px | 34205 | px |
| canalicular_length_density_per_px | 0.0334454 | px^-1 |
| junction_count | 701 | count |
| junction_density_per_px2 | 0.00068543 | px^-2 |
| lacuna_count | 14 | count |
| lacunae_per_px2 | 1.369e-05 | px^-2 |
| skeleton_component_count | 298 | count |

| | | |
|---|---|---|
| mean_component_length_px | 114.782 | px |
| median_component_length_px | 19 | px |
| field_length_density_w_per_px | 0.0378572 | px^-1 |
| field_density_without_flagged_per_px | 0.034249 | px^-1 |
| field_density_in_roi_per_px | | px^-1 |

Canalicular width, the new measure

| measure | value | units |
|---|---|---|
| width_mean_px | 6.9427 | px |
| width_median_px | 7.2111 | px |
| width_p10_px | 4 | px |
| width_p90_px | 10 | px |
| width_pixels_used | 32870 | count |

Canalicular width: 2 x the Euclidean distance transform of the canalicular mask at each skeleton pixel.
Measured only on skeleton pixels on the canalicular mask: the lacuna buffer and the
dilated vascular mask are excluded. The threads are about 2 to 6 px wide, so the width is close to the
pixel grid and quantised in steps of about 1 px, it moves with the threshold (a lower cut widens every
thread), and it has not been validated against any manual measurement.

Lacuna shape

| lacuna_id | area_px2 | major_axis_length_px | minor_axis_length_px | aspect_ratio | eccentricity | solidity | orientation_rad | centroid_row_px | centroid_col_px | on_border |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 510 | 36.0388 | 20.5481 | 1.7539 | 0.8215 | 0.8095 | -0.2657 | 13.3863 | 106.873 | True |
| 2 | 448 | 43.7126 | 19.8458 | 2.2026 | 0.891 | 0.6767 | 1.435 | 7.0045 | 363.067 | True |
| 3 | 1379 | 89.4664 | 21.3431 | 4.1918 | 0.9711 | 0.7971 | -1.5436 | 60.8564 | 471.593 | False |
| 4 | 1062 | 83.8665 | 19.2697 | 4.3522 | 0.9732 | 0.6923 | -1.1353 | 135.124 | 920.395 | False |
| 5 | 4132 | 136.42 | 39.4244 | 3.4603 | 0.9573 | 0.9008 | 1.5344 | 431.979 | 214.138 | False |
| 6 | 1032 | 65.9485 | 21.7201 | 3.0363 | 0.9442 | 0.8418 | -0.9507 | 441.636 | 703.582 | False |
| 7 | 2868 | 91.0398 | 40.6809 | 2.2379 | 0.8946 | 0.9166 | -1.49 | 534.719 | 477.113 | False |
| 8 | 980 | 60.8458 | 24.7024 | 2.4632 | 0.9139 | 0.7644 | -1.4596 | 557.819 | 348.526 | False |
| 9 | 2962 | 111.965 | 35.5084 | 3.1532 | 0.9484 | 0.917 | 1.4275 | 591.514 | 47.6516 | True |
| 10 | 2810 | 108.692 | 33.8251 | 3.2134 | 0.9503 | 0.9201 | 1.4475 | 714.117 | 172.09 | False |
| 11 | 1297 | 66.0375 | 26.8577 | 2.4588 | 0.9136 | 0.9134 | -1.189 | 725.743 | 851.213 | False |
| 12 | 1519 | 86.1566 | 25.1812 | 3.4215 | 0.9563 | 0.8388 | 1.3918 | 837.228 | 113.164 | False |
| 13 | 2083 | 63.4325 | 42.3049 | 1.4994 | 0.7451 | 0.9485 | -1.2565 | 905.171 | 752.206 | False |
| 14 | 2782 | 91.2948 | 39.6911 | 2.3001 | 0.9005 | 0.8762 | -1.4405 | 919.61 | 300.233 | False |

Network per lacuna

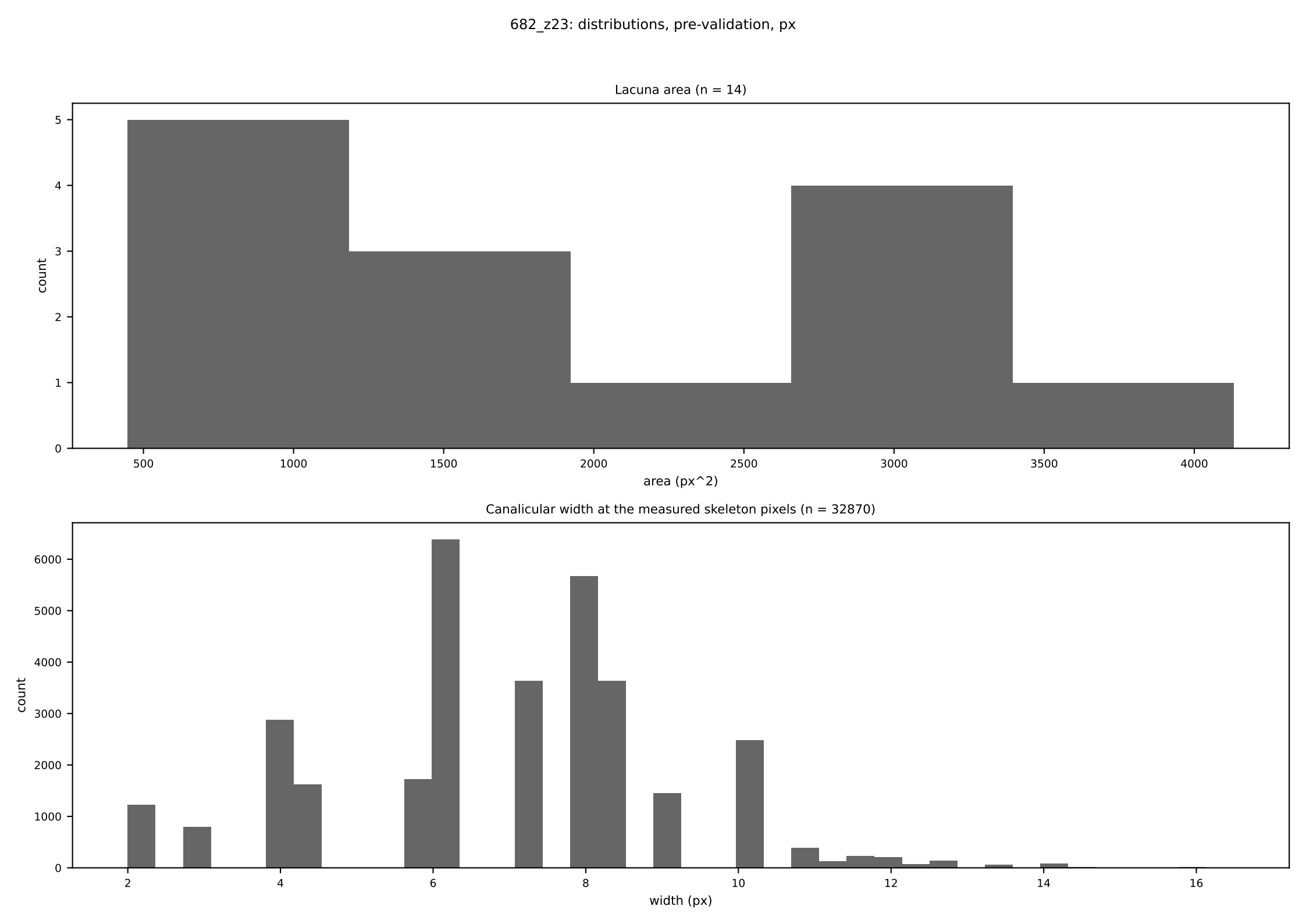
| lacuna_id | roots_count | ring_length_r30_px | ring_length_r60_px | owned_length_px | edge_count | mean_edge_length_px | ring_width_mean_r30_px |
|---|---|---|---|---|---|---|---|
| 1 | 3 | 96 | 222 | 2226.18 | 107 | 20.8054 | |
| 2 | 2 | 42 | 91 | 36.3137 | 4 | 9.0784 | |
| 3 | 8 | 326 | 736 | 1052.7 | 46 | 22.8847 | 6.7931 |
| 4 | 6 | 334 | 824 | 2781.32 | 135 | 20.6024 | 5.8974 |
| 5 | 9 | 344 | 906 | 2586.39 | 117 | 22.1059 | 6.5577 |
| 6 | 5 | 203 | 681 | 1286.96 | 71 | 18.1262 | 6.8732 |
| 7 | 8 | 325 | 804 | 4019.58 | 187 | 21.4951 | 6.1013 |
| 8 | 9 | 427 | 909 | 2843.46 | 118 | 24.0971 | 6.4156 |
| 9 | 6 | 230 | 651 | 1554.84 | 73 | 21.2992 | 6.6498 |
| 10 | 4 | 335 | 1099 | 1539.45 | 59 | 26.0925 | 6.5434 |
| 11 | 4 | 220 | 613 | 360.149 | 15 | 24.01 | 5.8424 |
| 12 | 6 | 285 | 835 | 1173.8 | 48 | 24.4542 | 5.5278 |
| 13 | 5 | 177 | 686 | 903.708 | 38 | 23.7818 | 6.2404 |
| 14 | 9 | 325 | 918 | 889.838 | 37 | 24.0497 | 5.9713 |

Normalised per lacuna

| lacuna_id | perimeter_px | ring_area_r30_px2 | ring_area_r60_px2 | in_frame_fraction_r30 | in_frame_fraction_r60 | ring_density_r30 | ring_density_r60 | roots_per_100px_perimeter |
|---|---|---|---|---|---|---|---|---|
| 1 | 105.154 | 3701 | 10161 | 0.6475 | 0.597 | 0.0259389 | 0.0218482 | 2.8529 |
| 2 | 132.847 | 3699 | 7686 | 0.6031 | 0.5656 | 0.0113544 | 0.0118397 | 1.5055 |
| 3 | 219.681 | 8976 | 20715 | 1 | 0.9689 | 0.0363191 | 0.0355298 | 3.6416 |
| 4 | 210.244 | 8511 | 22591 | 1 | 1 | 0.0392433 | 0.0364747 | 2.8538 |
| 5 | 335.421 | 12178 | 29932 | 1 | 1 | 0.0282477 | 0.0302686 | 2.6832 |
| 6 | 167.238 | 7485 | 20582 | 1 | 1 | 0.0271209 | 0.0330872 | 2.9898 |
| 7 | 243.522 | 9472 | 22056 | 1 | 1 | 0.0343117 | 0.0364527 | 3.2851 |
| 8 | 167.924 | 7349 | 17555 | 1 | 1 | 0.0581031 | 0.0517801 | 5.3596 |
| 9 | 282.894 | 8678 | 19332 | 0.813 | 0.7462 | 0.0265038 | 0.0336747 | 2.1209 |
| 10 | 263.48 | 10172 | 24552 | 1 | 1 | 0.0329335 | 0.0447621 | 1.5181 |
| 11 | 174.51 | 7570 | 20770 | 1 | 1 | 0.0290621 | 0.0295137 | 2.2921 |
| 12 | 210.167 | 8646 | 22189 | 1 | 1 | 0.0329632 | 0.0376313 | 2.8549 |
| 13 | 185.823 | 7980 | 21610 | 1 | 1 | 0.0221805 | 0.0317446 | 2.6907 |
| 14 | 259.279 | 9669 | 24825 | 1 | 1 | 0.0336126 | 0.0369789 | 3.4712 |

Network v2 per lacuna

| lacuna_id | ring_attached_length_r30_px | ring_attached_length_r60_px | ring_length_w_r30_px | ring_length_w_r60_px | sholl_crossings_r10 | sholl_crossings_r20 | sholl_crossings_r30 |
|---|---|---|---|---|---|---|---|
| 1 | 81 | 153 | 103.426 | 244.238 | 3 | 2 | 4 |
| 2 | 38 | 38 | 43.1421 | 98.7696 | 1 | 0 | 0 |
| 3 | 186 | 446 | 363.078 | 813.47 | 7 | 12 | 13 |
| 4 | 284 | 624 | 381.12 | 948.092 | 9 | 11 | 10 |
| 5 | 314 | 607 | 386.12 | 1011.64 | 9 | 12 | 12 |
| 6 | 163 | 324 | 227.409 | 782.21 | 6 | 7 | 6 |
| 7 | 263 | 505 | 354.451 | 900.139 | 8 | 11 | 11 |
| 8 | 404 | 806 | 479.605 | 1014.38 | 11 | 14 | 15 |
| 9 | 144 | 336 | 251.51 | 728.186 | 5 | 10 | 9 |
| 10 | 221 | 658 | 377.664 | 1230.13 | 6 | 15 | 17 |
| 11 | 125 | 207 | 247.48 | 707.411 | 5 | 10 | 10 |
| 12 | 220 | 547 | 330.191 | 946.879 | 7 | 7 | 18 |
| 13 | 139 | 376 | 201.167 | 786.696 | 5 | 7 | 11 |
| 14 | 304 | 615 | 363.664 | 1032.02 | 10 | 12 | 14 |


682_z23: distributions, pre-validation, px
Lacuna area (n = 14)
count
area (px^2)
0
1
2
3
4
5
500
1000
1500
2000
2500
3000
3500
4000
Canalicular width at the measured skeleton pixels (n = 32870)
count
width (px)
0
1000
2000
3000
4000
5000
6000
2
4
6
8
10
12
14
16